

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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03/13/25

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R2401018

Order Instituting Rulemaking to Establish
Energization Timelines.

Rulemaking 24-01-018
(Filed January 25, 2024)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON
ADMINISTRATIVE LAW JUDGE'S RULING CLARIFYING NEXT STEPS FOR
FLEXIBLE SERVICE CONNECTIONS, MODIFYING PHASE 2 SCHEDULE, AND
REQUESTING PARTY COMMENTS**

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Date: March 13, 2025

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments to the *Administrative Law Judge’s Ruling Clarifying Next Steps for Flexible Service Connections, Modifying Phase 2 Schedule, and Requesting Party Comments* (“Ruling”) filed by Administrative Law Judge (“ALJ”) Justin Regnier on February 7, 2025, *Email Ruling Correcting Error in the February 7, 2025, Administrative Law Judge’s Ruling Re: Filing Date For Reply Comments* filed by ALJ Carolyn Sisto and ALJ Justin Regnier on February 10, 2025, and *E-Mail Ruling Granting Schedule Amendment* filed by ALJ Justin Regnier on February 20, 2025.

I. INTRODUCTION.

VGIC commends the Commission for moving forward with the next steps for leveraging Flexible Service Connections (“FSCs”) in California. As discussed by VGIC in previous comments, FSCs have the potential to greatly accelerate energization timelines to allow for partial energization of sites and unlock significant ratepayer savings in the long term through distribution infrastructure upgrade deferral. VGIC supports the Commission’s plans to move forward with “how to standardize FSC in simple terms that can be implemented in the very near term to avoid

energization delays triggered by upstream capacity needs on the distribution system.”¹ This proceeding will discuss the implementation of static and pre-programmed Limited Load Profiles (“LLPs”). VGIC agrees with this near-term, but we recommend the Commission also consider some elements of flexibility that will be important for the longer-term implementation of FSCs and Flexible Service Connection Agreements (“FSCA”). We respond to the questions outlined in the Ruling below.

II. RULING QUESTIONS.

Pilot and Process Learnings

1. What learnings from past and ongoing large IOU FSC pilots and processes can inform the design of LLP?

In California there are two FSC pilots that VGIC is aware of: PG&E’s Flex Connect and SCE’s Load Control Management Systems (“LCMS”). Despite these pilots being in place for over a year, there appears to be, to VGIC’s surprise, very limited publicly available data filed with the Commission or available on the IOUs’ websites from which VGIC or other parties may draw learnings. At present, this finding alone suggests a clear learning that forthcoming Commission direction must leverage: **FSC implementation progress must be reported to the Commission and public stakeholders in a format and consistency sufficient to inform future pilot, tariff, and/or program development.** VGIC is aware of only one instance in which PG&E and SCE were asked to provide information on the number of sites using FSC, in response to an ALJ ruling asking, “How many EV charging sites (including capacity), that are either operational or under development, are piloting or using a form of flexible connection

¹ Ruling at 4.

agreement that relies on power import limiting schedules?” PG&E reported 60 customers on static load limits and that additional customers were planned to join the dynamic pilot in 2024, and SCE reported that 4 customers were in the LCMS pilot but that they were in discussion with 10 more customers at the time.² However, this data is now more than 6 months old and out of date.

VGIC strongly recommends the Commission direct the IOUs to report quarterly or annually on the number of sites awaiting grid connection, the number of sites pursuing FSC/FSCA, the type of FSC schedule used, the type of FSC technology used, and related details that demonstrate the timeline for energization under an FSC path, should the customer and/or utility choose to pursue it. **VGIC reiterates its proposed template for IOU reporting on FSC, detailed in this proceeding in its May 3, 2024 Comments of VGIC on Phase 1 Scoping Memo Issues, and copied in the figure below.**³

VGIC Comments Section II.A subsection:	Project Specifics							Flexible Service Connection						Milestones pursuant to Commission direction in R.24-01-018				
	(i) - Project Specifics			(ii) - Project Specifics			(iii) - Tariff	(iv)(a)-(b) - Utility-Proposed Process		(iv)(c)-(d) - Customer-Proposed Process		(iv) - Type and Tech		(iv) - Milestones				
	Project ID	Utility	ZIP Code	Request Type	Project Status (Current Phase)	Size (kW, Net, Nameplate, Rating)	Tariff	Flex Connect Proposed by Utility?	Flex Connected Accepted by Customer?	Flex Connect Proposed by Customer?	Flex Connect Accepted by Utility?	Flex Connect Type	Flex Connect Tech	Date of Initial Energization Request	Days from submission of Initial Energization Request to Applicant Final Submission (AFS)	Days from Applicant Final Submission to [milestone 3]	Days from [milestone 4] to customer service transformer energization	Notes
		PG&E		New Load	APPROVED	100	R15 & R16	Yes	Yes	No	N/A	Single-Value	PCS	1/1/2025	8	10	12	
		SCE		New Load	SUBMITTED	60	R29 & R21	No	N/A	Yes	Yes	288-Value	Storage	45658	12	16	10	[R21 Project ID X]

VGIC is disappointed that the time has come to learn from PG&E and SCE’s FSC pilots, yet parties have limited ability to understand how those pilots have yielded benefits as the IOUs were not required by the Commission to report out on these findings. This is out of step with the Commission’s standard course of action for new IOU pilots, tariffs, and programs.

² See R.21-06-017, Pacific Gas and Electric Company (U 39 E) Comments Responding to Questions Contained in Attachment 3 to Administrative Law Judge’s Ruling Providing Two Working Group Reports and Directing Responses to Questions on Reports at p.10,

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K116/535116411.PDF>; and R.21-06-017, Southern California Edison Company’s (U 338-E) Opening Comments on Track 3 Working Group Reports at p.15, <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K173/535173390.PDF>.

³ R.24-01-018. Comments of VGIC on Phase 1 Scoping Memo Issues.

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M531/K294/531294238.PDF>

Through PG&E's and SCE's responses to the ALJ Ruling filed in R.21-06-017 on May 29, 2024, and through interviews VGIC has since conducted with IOU management and VGIC members, VGIC has been able to learn more about the implementation strategies, even if we do not understand the effects these strategies are having on energization timelines and distribution system upgrade costs. Based on the July 2024 responses to the ALJ Ruling in R.21-06-017, PG&E and SCE offer simple four- and six-value schedules to each participant based on the customer's needs and the potential impact at that location on the grid.⁴ VGIC also understands that PG&E now offers a more dynamic approach, integrating sites into its DERMS platform and dispatching load reduction as needed. VGIC recommends that, at a minimum, the Commission extend the current FSC schedules (i.e., simple 4- or 6-value schedules as well as PG&E's optional dynamic integrated approach) to the initial LLP offerings adopted in this proceeding. Additionally, the existing FSCA language, e.g., SCE's Form 16-332, could serve as a rough basis for more permanent customer agreements, although VGIC offers more detailed recommendations in its below response to ALJ Ruling Question 12.

Notably, only PG&E and SCE offer FSC approaches. To create consistency and preempt the anticipated growth of service requests due to transportation electrification in constrained locations on the distribution grid, VGIC recommends the Commission direct all three large IOUs (i.e., including SDG&E) to offer scalable FSC options. In recognition of the differences in investments and IT capability across the three IOUs, namely that PG&E appears to have DERMS capability to implement dynamic FSC that the other IOUs do not, VGIC recommends

⁴ R.21-06-017, Pacific Gas and Electric Company (U 39 E) Comments Responding to Questions Contained in Attachment 3 to Administrative Law Judge's Ruling Providing Two Working Group Reports and Directing Responses to Questions on Reports at p.10, <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K116/535116411.PDF>; and R.21-06-017, Southern California Edison Company's (U 338-E) Opening Comments on Track 3 Working Group Reports at p.15, <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K173/535173390.PDF>

SDG&E use the simpler four- or six-value approach to FSC schedules. Lastly, VGIC strongly recommends the Commission direct all three IOUs to make FSC options available to all customers, not just large load customers. Notably, certain multi-family housing sites, smaller workplaces and fleet depots, and residential customers could also support both faster energization and distribution system upgrade deferral by signing an FSCA, if one were made available to them. Considering the looming \$50 billion in distribution system upgrade costs found by the Commission's consultant in R.21-06-017, VGIC believes offering scalable FSC options to all customers represents a no-regrets path forward to help mitigate this overwhelming potential ratepayer cost.⁵

2. Are there FSC processes in other jurisdictions whose learnings should be considered? If so, what are those jurisdictions and the associated learnings?

Based on VGIC's engagements in New York, Massachusetts, Colorado, Michigan, Illinois, Texas, and Maryland, we understand that all large IOU FSC processes outside of California are entirely *ad-hoc* and, generally, poorly defined. In other words, we are not aware of any standardized, scalable processes in the US that California can look to for learnings. The conventional approach to exploring flexibility within a new or upgraded service request relies on informal information exchange between the site developer and utility, and the content, frequency, and platform for that information exchanged is not standardized, but dependent on conversations between individuals at each organization. This approach does not scale well as it is resource intensive and, for site developers, represents a significant source of uncertainty.

⁵ Kevala prepared for CPUC. *Electrification Impacts Study Part I: Bottom-Up Load Forecasting and System-Level Electrification Impacts Cost Estimates*. May 9, 2023. https://cdn.prod.website-files.com/62a236e9692c48cff36898da/6462917ab8a790b6b85f5fbb_CPUC%20Kevala%20EIS%20Part%201.pdf

While it has not defined a scalable *process*, New York has authorized approximately \$25 million in funding specifically targeted to support the deployment of power control systems (“PCS”) and related technologies needed to enable FSC. The New York utilities currently administer this Load Technology Management Incentive Program (“LMTIP”), clearly detailing eligible PCS technologies and required standards.⁶ VGIC recommends the Commission consider a similar incentive approach to share the system cost savings of FSC between ratepayers and participating customers that incur PCS installation and maintenance costs. We provide additional detail on this recommendation below in response to ALJ Ruling Question 14.

Utilities elsewhere in the world do offer standardized, scalable FSC processes. Lessons learned and technical details can be found in the US DOE’s July 2024 *Flexible DER & EV Connections*.⁷ In Australia, the focus has been on dynamic operating envelopes for DERs, in which smart inverter functions have provided value through the use of ADMS and DERMS platforms. In the context of the Ruling’s questions regarding FSC, this is of limited use given not all sites seeking FSC will have smart inverters.

3. What is an appropriate temporal granularity for LLP schedules? For example, will the simple monthly or seasonal import-limit granularity utilized within Limited Generation Profiles (LGP) be sufficient for LLP schedules? If not, why not?

VGIC suggests providing a handful of different LLP options with different levels of temporal granularity, including:

A. 4- or 6-value structure offered in SCE and PG&E’s current FSC pilots.

⁶ See Load Management Technology Incentive Program Overview: <https://jointutilitiesofny.org/ev/lmtip>

⁷ US DOE. *Flexible DER & EV Connections*. July 2024. <https://www.energy.gov/sites/default/files/2024-08/Flexible%20DER%20%20EV%20Connections%20July%202024.pdf>

B. 288-value schedules based on the Limited Generation Profiles (“LGP”) schedules authorized in R.17-07-007.

C. More granular and dynamic schedules as also offered in PG&E’s Flex Connect pilot.

At this early stage, it is unclear whether there is one optimal LLP granularity level, and there are important tradeoffs to consider. Additional granularity can provide more value to certain customers, especially if meaningful import constraints are isolated to only a few hours of the day in a few months. At the same time, more complex LLP schedules can be harder to understand and implement for certain customers, e.g., shared heavy-duty truck charging sites. VGIC recommends the Commission leverage a combination of the current FSC pilot schedules and the work done to develop the LGPs to ensure there are several options offered to customers, considering the needs of customers are incredibly diverse. This avoids “reinventing the wheel” and instead leverages existing methodologies in support of achieving faster Commission approval, IOU implementation, and market response. **VGIC proposes the Commission ultimately direct IOUs offer all three of the above approaches as options for customers to choose from.**

Since all utilities offer 24-value LGPs under Rule 21, VGIC anticipates 24-value LLPs to be relatively easy to implement across all three IOUs. PG&E already offers both 4-value static and dynamic approaches under its Flex Connect, and SCE already offers a 6-value static schedule.⁸ This methodology is worth considering going forward given its simplicity and ease of

⁸ R.21-06-017, Pacific Gas and Electric Company (U 39 E) Comments Responding to Questions Contained in Attachment 3 to Administrative Law Judge’s Ruling Providing Two Working Group Reports and Directing Responses to Questions on Reports at p.10, <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K116/535116411.PDF>; and R.21-06-017, Southern California Edison Company’s (U 338-E) Opening Comments on Track 3 Working Group Reports at p.15, <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M535/K173/535173390.PDF>

understanding for customers. Meanwhile, more granular LLP schedules, such as the 24-value or fully dynamic approaches, would provide additional import ability. For many circuits that only face constraints in shorter and/or fewer periods across the day or year, more granular schedules will allow customers to maximize their import ability during times where there are no constraints. For example, PG&E has shown an illustrative example of a customer energization where a constraint was found only during 3 months during 3-11PM. In this case, PG&E would be able to fully serve load for ~90% of hours during the year and only need to put a constraint on the other 10% of hours.⁹ If this customer were only able to choose a 4-value LLP with equal seasons and peak periods, they would need to constrain their imports for 25% of the year. **For this reason, VGIC strongly urges the Commission to adopt more than one optional schedule structure.**

If standard LLP schedules are adopted, VGIC encourages the Commission to adopt options for more granular 24-value LLP configurations. A good starting point for creating these more granular configurations is to consider the schedules developed for the LGPs. These are:

- 24-Hourly: One value for each hour of the day across all 12 months
- Block: One value for each pairing of a seasonal block and an hourly block. Divides the year into four blocks of 3-month intervals and each day into six blocks of 4-hour intervals.
- 18-23-Fixed: One value for each of two different block intervals each day: 6pm-12am and 12am-6pm, uniquely for each month of the year.

⁹ R.23-12-008. Vehicle-Grid Integration Forum Workshop Report Filed by Southern California Edison Company (U 338-E), San Diego Gas & Electric Company (U 902 E), and Pacific Gas and Electric Company (U 39 E). <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M532/K262/532262533.PDF>

- Note: The 18-23-fixed schedule is well suited to accommodate exports and was shown to maximize power output from LGP projects while minimizing thermal violations. However, most exporting resources are solar projects that provide significant exports during the day and no exports at night. This specific configuration may or may not be the most optimal schedule for electric imports. Additional analysis would be helpful to confirm which specific two-block schedule is expected to be the most utilized in an LLP construct. However, VGIC does support the overall structure of this configuration: two-time block intervals that are unique for each month.

A menu of different options for customers will provide the IOUs and customers with flexibility to consider which options are best for their expected load profiles and the unique constraints and considerations for their specific circuit.

VGIC also encourages the Commission to consider more granular LLP configurations, including a 288-value configuration with a unique value for each hour in each month (12 months x 24 hours). Significant value can be unlocked from maximizing the import potential throughout the year, especially if LLPs are used for a prolonged period or as a permanent energization configuration. There were discussions on LGPs about these complex configurations but raised issues around the potential for safety violations. It is unclear to VGIC if these same safety concerns for granular schedules exist on the load side. If significant safety concerns exist for the adoption of a 288-value schedule currently, VGIC suggests adopting a menu of simpler options at this stage and revisiting the issue as more data is gathered on customers usage of LLPs and as more complex FSC structures are adopted in the High DER proceeding.

4. What elements of LGP adopted in R.17-07-007 and LGP Resolutions should be adapted for development of LLPs? In what ways can implementation of LLP employ approaches from the LGP process? In what ways should LLP design and implementation differ from LGP?

The Commission should leverage as many learnings from the LGPs as possible, given the extensive amount of work conducted by stakeholders in the development of those profiles. VGIC is strongly supportive of the schedule to adopt LLPs and near-term FSCs put forward by the Commission in this Ruling that includes a Proposed Decision being issued in Q3 of 2025. Adoption of widespread FSC by the Commission is needed now to address the existing and worsening energization backlogs. At the same time, this means that we cannot repeat the more than 5-year-long process it took to finalize the LGPs in the Rule 21 proceeding.

VGIC believes that the LGP structures should be considered as jumping off points. For example, the Commission should consider moving forward with 24-value structures, as these were found to be optimal for the LGPs.

5. How can Load ICA results and data inform and enable LLPs? Is there another existing means to inform and enable LLP other than Load ICA?

VGIC believes that the Load ICA should be improved and leveraged to develop and enable LLPs, especially as this option becomes more widespread. Currently, it is VGIC's understanding that IOUs utilize their own internal data that is not included in the ICA to develop the LLPs utilized in today's FSC pilots. This is a feasible method when there are only a few customers participating in a pilot program. However, it will likely become much more time intensive and difficult to conduct these studies as the number of energization requests triggering upgrades increases and as FSCs become more popular. Leveraging a granular load ICA to develop an LLP will be critical to

the widespread adoption of FSCs and will help customers understand their own FSC options without having to add work to utility staff. Ultimately, as was adopted for the LGPs for the generation hosting capacity maps, the Commission and utilities should work towards creating the functionality for customers to develop LLPs in the load ICA.

However, there are some baseline errors and constrained functionality that prevent LLPs from being able to be developed from the current version of the Load ICA maps. PG&E is currently implementing workflows to use the Load ICA maps in internal energization processes to identify if an energization application will face capacity constraints.¹⁰ However, the other utilities' ICA maps are not currently reliable enough to be utilized in the same manner, with the Commission noting "SCE's participation in this activity [(utilizing the ICA maps for internal energization screenings)] would need to be delayed until they improve the accuracy of their ICA results but [Commission staff] suggest SCE should begin planning for the work."¹¹ SCE has stated that they would not be able to incorporate the Load ICA into energization screening processes until at least Q3 2026.¹² Until these fundamental issues of ICA accuracy are addressed, customers will not be able to develop meaningful LLPs utilizing these tools. When LLPs are adopted in this proceeding, the Commission should direct the IOUs to ensure Load ICA accuracy and granularity as quickly as possible, e.g. in Q1 2026.

Equipment Requirements and Certification

6. To implement static LLPs what device(s) are required for customers to install to control load and prevent power consumption (i.e., imports) from exceeding the scheduled LLP? Is PCS sufficient for this task?

¹⁰ D.24-10-030 at p.174.

¹¹ Ibid at p.175.

¹² See SCE Advice Letter 5445-E.

Yes, based on VGIC's understanding, PCS is sufficient to prevent imports from exceeding LLP limits.

7. Is a PCS certified to UL 3141 Issue 2,4 sufficient to operationalize static LLPs? If no, why not, and are there alternative existing standards or equipment the Commission should consider for LLP participation?

VGIC believes that utilizing PCS certified to the UL 3141 is sufficient to operationalize static LLPs. The second edition of UL 3141 published in October 2024 includes parameters for scheduling Power Import Limits.

8. When do stakeholders expect UL 3141 Issue 2 certified equipment to be readily available to support LLP?

As VGIC understands it, technology providers are working toward UL 3141 Issue 2 certification, although the exact timeline is unclear at present. Given the fast-changing dynamics in this space, VGIC encourages the Commission and Energy Division staff to meet with individual stakeholders to confirm timelines ahead of issuing its proposed decision to establish equipment requirements.

9. Is there a size threshold (e.g., absolute [MW] or relative [percent of total feeder rating] site capacity) that necessitates equipment commissioning or telemetry? If so, what is that threshold and what need does it raise?

VGIC understands the desire to ensure that there are safe operations under the LLPs, but we caution against telemetry requirements for static (i.e., "set-and-forget") LLP implementation. Telemetry equipment, especially utility-grade telemetry, can be extremely expensive and add significant complexity to a project. Particularly for situations in which FSCs with static, "set-and-

forget” LLPs are being used as a bridge solution for only a few months or years, telemetry should not be required. Some level of communication capability is needed for dynamic LLPs or FSCs that are currently being implemented by PG&E in the Flex Connect Pilot, and communications equipment requirements have already been outlined by PG&E. VGIC recommends that these requirements for dynamic FSCs adopted as near-term bridging solutions continue to be discussed with the utilities on an individual basis. VGIC believes that issues surrounding standardizing possible telemetry needs for active communication or more complex non-firm FSCs should be taken up in Track 3 of the High DER proceeding, where long-term FSCs and non-firm capacity is being discussed.

Energization Queue and Circumstances for LLP

10. Under what circumstances should a customer be eligible for FSCAs? What type or class of customer should be eligible for FSCAs and why? Should FSCAs be reserved for when a circuit is constrained? Should customers be able to elect to engage in an FSCAs on an unconstrained circuit to aid in creating circuit capacity for additional customers?

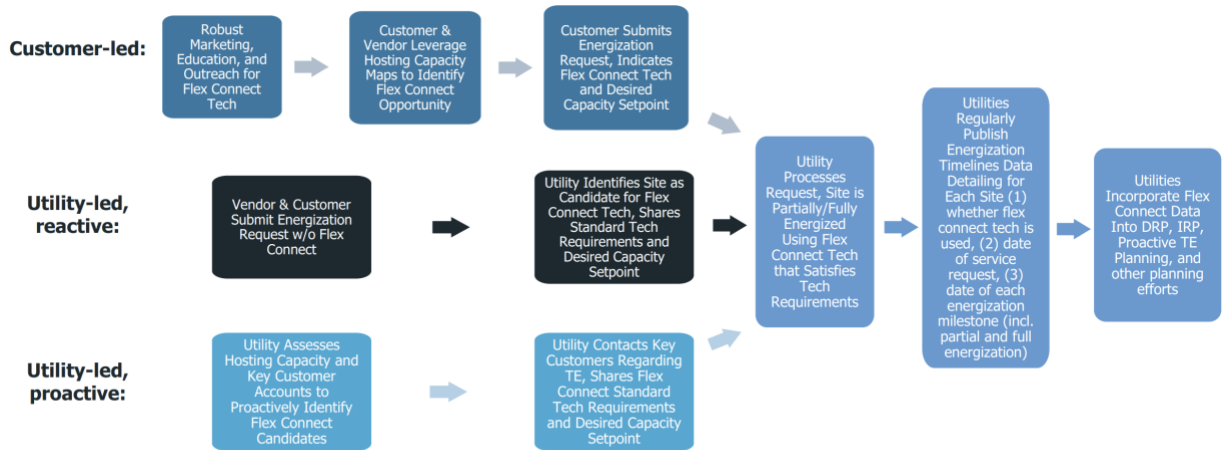
VGIC will focus on FSCAs for EV customers, which we believe should be eligible for any FSCA structure or program the Commission adopts. Fundamentally, VGIC believes that all EV customers should be *eligible* for FSCAs, especially in the long-term, but that FSC participation should always be voluntary. There are significant ratepayer benefits to having customers voluntarily engage in FSCAs, even in unconstrained circuits where election of FSCAs can aid in creating circuit capacity for additional customers in the future. VGIC discusses this in more detail in response to the Ruling on FSCs in the High DER proceeding (R.21-06-017). However, there aren’t many incentives for customers to voluntarily agree to an FSC given current rules. At this time, the Commission should focus its FSC design efforts on customers that are likely to adopt

FSCAs today, which are customers on constrained circuits, however, the Commission should not necessarily preclude customers on unconstrained circuits from electing FSCAs.

There are two incentives that push customers towards adopting FSCs: the potential to energize more quickly and the potential to save money. If there are no constraints on a circuit, customers are unlikely to save significant amounts of time or money by entering an FSCA. Under Rule 15 and 16, customers are exposed to some elements of the costs they are creating on the grid if upgrades exceed their allowances. However, for many EV customers leveraging Rule 29/45, additional costs are covered. Rule 29/45 has been critical to EV adoption, especially in the medium- and heavy-duty sectors that require significantly more power but also provide some of the greatest greenhouse gas and air quality benefits by electrifying. If customers had to pay their full upgrade costs, it would be detrimental to transportation electrification. However, because of this, EV customers are more likely to be driven to adopt FSCs in order to accelerate their energization timelines, as opposed to electing FSC for cost reasons.

This does not necessarily affect customer *eligibility* for FSCAs, but VGIC does believe that this should impact marketing and education for FSCAs, as well as active outreach by the utilities on FSC options. During the 2024 VGI Forum, hosted pursuant to D.22-11-040, VGIC outlined three potential pathways for customers to enter into an FSCA: a customer-led path, a utility-led reactive path, and a utility-led proactive path. See details on these pathways below, as filed in R.23-12-008¹³:

¹³ R.23-12-008. Vehicle-Grid Integration Forum Workshop Report Filed by Southern California Edison Company (U 338-E), San Diego Gas & Electric Company (U 902 E), and Pacific Gas and Electric Company (U 39 E). May 21, 2024. <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M532/K262/532262533.PDF>



For the utility-led pathways, EV customers that will face long energization timelines will likely be most interested in FSCAs. Therefore, utilities that receive EV energization applications should react accordingly to provide FSC options if they can help a customer achieve a shorter timeline. Utilities should also consider how to proactively outreach to customers on constrained or nearly constrained circuits they know will electrify their fleets to share these FSC options with them.

11. How should the Utilities address multiple requests for FSCAs on the same circuit?

At this stage, it is still unclear which way to handle multiple FSCAs on the same circuit is best for ratepayers and FSC participants. VGIC understands that SCE is currently only allowing one customer on a given circuit to participate in its LCMS pilot. Under the current LCMS pilot, the customer that engages in the FSCA is allocated all available firm headroom on that circuit while the distribution upgrade is being constructed, so if another customer requests to energize new load on that circuit, there is no extra capacity until the upgrade is completed. Since SCE reactively selects LCMS participants from its energization queue, VGIC believes it is not currently possible for two FSC energization applications on the same constrained circuit to have to share

capacity in any way. As FSC becomes a scalable, standardized offering, it is unclear how common this will become. At this time, moving forward with a “first-come, first-serve” structure for bridging solutions is likely most feasible to implement. **Here, VGIC reiterates its recommendation for clear reporting requirements of existing and forthcoming FSC efforts, so the Commission can devise an effective capacity allocation methodology based on real data provided by the IOUs.**

If more permanent FSC structures are adopted with non-firm capacity allocations, then the Commission should consider schedules for reassessment or other frameworks for allocating capacity between multiple customers. VGIC is aware of discussions regarding a “use it or lose it” approach for firm or non-firm import or export capacity, where customers that are not utilizing their full capacity reservations have that capacity reallocated to other customers. VGIC looks forward to considering these frameworks for long-term FSC implementation in Track 3 of the High DER proceeding. For the focus of this proceeding, i.e., static LLPs solely utilized as bridging solutions, these frameworks are likely too complex for immediate implementation.

Tariffs, Rules, Agreements, and Forms

12. What are the applicable tariffs, rules, agreements, and forms that may need modification? What should those modifications be? Is there a need for differences between different IOUs’ versions of these documents? If so, what is this need based upon?

In order to implement FSCs, an FSC agreement (FSCA) will need to be developed for customers to sign. This will outline the FSC schedule and terms they are agreeing to and will also ensure that customers have installed the necessary PCS or other equipment to adhere to the LLP. SCE’s Form 16-332 for the LCMS pilot provides an example of what this type of agreement could look like. However, there are some parameters of SCE’s existing Form 16-332 which greatly risks

a positive customer experience, including the ability to change LCMS parameters at any time with customers needing to adhere to new limits within 24 hours or less.¹⁴ A permanent framework should include more specificity around whether and how the IOUs can change firm import limits and should consider better protections for customers. Without these customer protections, VGIC is concerned over participant backlash and limitations on FSC election, jeopardizing both energization timelines and the opportunities for ratepayer cost relief through distribution upgrade deferral.

Modifications to IOU energization rules are also needed for implementation of LLPs. For example, SCE's Rule 16 and Rule 29 currently note the following: "Installation of automated load management software pursuant to the LCMS Pilot and passive, non-electrically connected monitoring devices on or near the meter by non-SCE personnel is permitted."¹⁵ This reference to automated load management and specific LCMS pilot would likely need to be changed to "Installation of power control systems." In PG&E's Rule 2, there are references that a customer's connected load will be deemed as "the sum of the rated capacities of all of the Customer's electric utilization equipment that is served through one metering point and that may be operated at the same time."¹⁶ This may also need to be modified to acknowledge the use of power control systems and/or the election to electrify less than the sum of the rated capacities during certain hours, as agreed upon within the FSCA.

The Commission should also implement changes to energization rules that allow for flexibility in the FSC structures that could be adopted in the future. The High DER proceeding (R.21-06-017) is considering the long-term implementation of FSCs that could include more flexible

¹⁴ SCE Advice Letter 5138-E.

¹⁵ SCE Rule 16 and Rule 29 at 12. Service Connections.

¹⁶ PG&E Rule 2 at H.1. Connected Load Ratings.

structures and non-firm capacity. Any language for energization rules that is adopted in response to the ALJ Ruling in this Energization Timelines proceeding (R.24-01-018) should be flexible enough to allow for elements such as more granular LLP schedules or non-firm capacity agreements and constructs that may emerge from the High DER Proceeding (R.21-06-017).

13. Are there any unique considerations that must be included in modifications to rules, agreements, and/or forms (e.g., such as for onsite generation, electric vehicles, or emergency conditions)?

The Commission should consider how to combine FSCs with Rule 21 and onsite generation and energy storage. VGIC anticipates that EV customers will explore meeting their charging needs during constrained hours by installing onsite DERs. This can allow for additional load to be served without requiring grid imports. PCS at the point of common coupling can ensure that import and export limits are not exceeded, but any parameters around having both load and generation behind-the-meter will need to be specified.

14. Are there other steps needed to implement FSCs? If so, what are these steps?

VGIC reiterates that successful, sustainable, and scalable implementation of FSC depends on the Commission's ability to track the outcomes of FSC. As such, VGIC urges the Commission to adopt IOU reporting requirements. VGIC acknowledges that both IOUs and the Commission have limited resources and that reporting requirements can be viewed as an undue burden or added "red tape." However, considering both nascency of scalable FSC frameworks globally and the criticality of FSC for California's electrification efforts, this is one domain in which public data reporting will be immensely informative for further policy development by the Commission.

Lastly, VGIC notes that customers electing FSC in response to grid constraints are enabling the deferral of costly distribution upgrades. Due to the fact that these costs are largely socialized

among all ratepayers, for example under Rule 29/45, and considering the time value of money that underpins the distribution upgrade deferral value proposition, VGIC notes that customers with FSCAs will materially reduce system costs compared to a non-FSC approach. While ratepayers should be entitled to the majority of those avoided system costs, the customer electing the FSC will incur upfront and/or ongoing costs by installing and maintaining an eligible PCS. Depending on the PCS, this may include costly hardware like gateways, relays, and/or stationary energy storage as well as software solutions, which may be paid for monthly or annually. With these costs and benefits in mind, VGIC respectfully urges the Commission to adopt a “shared savings” model in which FSC customers receive a share of the distribution system upgrade deferral value. In New York, the Public Service Commission has authorized approximately \$25 million in funding specifically targeted to support the deployment of technologies that underpin FSC. The New York utilities currently implement this Load Technology Management Incentive Program, detailing eligible PCS technologies and required standards.¹⁷ VGIC recommends the Commission adopt a similar technology incentive program to offset the customer costs of electing FSC and installing/maintaining PCS to unlock greater anticipated ratepayer benefits of (a) accelerated electrification and (b) distribution upgrade deferral. Given the affordability crisis and Governor Newsom’s Executive Order N-5-24, the total costs of this incentive program should not outweigh the ratepayer benefit from scaling FSC.

III. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Ruling. We look forward to further collaboration with the Commission and stakeholders on this initiative.

¹⁷ See Load Management Technology Incentive Program Overview: <https://jointutilitiesofny.org/ev/lmtip>

Respectfully submitted,

/s/ Zach Woogen

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VEHICLE-GRID INTEGRATION COUNCIL

Date: March 13, 2025